Physics Department
Fisika Departement

University of Pretoria

Semester Test 2 FSK176 *Semestertoets 2*

6 October/Oktober 2010

Surname and Initials: Memo____________________ Student Number: ____________________
Van en voorletters: *Studentenommer:*

Signature: __________________________
Handtekening:

Time : 2½ hours Total : 80 Maximum
Tyd: 2½ ure *Totaal: 80 Maksimum*

Answer all questions.	*Beantwoord alle vrae.*
Free-body diagrams are essential.	*Vry-liggaamsdiagramme is noodsaaklik.*
Programmable calculators may NOT be used.	*Programeerbare sakrekenaars MAG NIE gebruik word nie.*
No textbooks allowed.	*Geen handboeke word toegelaat.*
A formula sheet is given below.	*'n Formuleblad word gegee.*
All test and examination regulations of the University of Pretoria are applicable.	*Alle toets- en eksamenregulasies van die Universiteit van Pretoria is van toepassing.*

Formulas / Formules

See next page *Sien volgende bladsy*

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SEMESTER TEST 2 **FSK 176** **SEMESTERTOETS 2**

6 October/Oktober 2010

Surname and Initials:____________________________ Student Number: ____________________
Van en voorletters: *Studentenommer:*

Signature: __________________________ Venue:_____________________
Handtekening: *Lokaal:*

Formula/ Constant Sheet **Formule- en konstante bladsy**

Circle: Circumference $=2\pi r$ = omtrek

Area = πr^2

Sphere: Area = $4\pi r^2$

Volume = $^4/_3 \pi r^3$

Cylinder / Silinder

Area = $2(\pi r^2) + 2\pi r h$

Volume = $\pi r^2 h$

Trapesium: Area = $½(a+b)h$

$$\vec{a}\cdot\vec{b} = ab\cos\phi$$

$$\vec{a}\cdot\vec{b} = a_x b_x + a_y b_y + a_z b_z$$

$$\vec{a}\times\vec{b} = \left(a_y b_z - b_y a_z\right)\hat{i} + \left(a_z b_x - b_z a_x\right)\hat{j} + \left(a_x b_y - b_x a_y\right)\hat{k}$$

$$x = \frac{-b\pm\sqrt{b^2-4ac}}{2a}$$

$$v_{avg} = ½\left(v_0 + v\right)$$

$$v = v_0 + at$$

$$x = x_0 + v_0 t + ½at^2$$

$$x = x_0 + v_{avg} t$$

$$v^2 = v_o^2 + 2a\left(x - x_0\right)$$

$$x - x_0 = ½\left(v_0 + v\right)t$$

$$x - x_0 = vt - ½at^2$$

$$a = \frac{v^2}{r}.$$

g = 9,8 m/s^2

$$h = \frac{v_0^2\sin^2\theta}{2g}$$

$$R = \frac{v_0^2}{g}\sin 2\theta_0$$

$$T = \frac{2v_0\sin\theta}{g}$$

$$f = \mu N$$

$$D = ½C\rho A v^2$$

$$W = \vec{F}\cdot\vec{d} = Fd\cos\phi$$

$$F = -kx$$

$$W = -½kx^2$$

$$P_{avg} = \frac{W}{\Delta t}$$

$$P = \frac{dW}{dt}$$

$$P = \vec{F}\cdot\vec{v}$$

$$U = ½kx^2$$

$$x_{com} = \frac{1}{M}\sum_{i=1}^{n} m_i x_i$$

$$\vec{r}_{com} = \frac{1}{M}\sum_{i=1}^{n} m_i \vec{r}_i$$

$$\vec{p} = m\vec{v}$$

$$\vec{F}_{net} = \frac{d\vec{p}}{dt}$$

$$\Delta\vec{p} = \vec{J}$$

$$\vec{J} = \int_{t_i}^{t_f} F\left(t\right)dt$$

$$J = F_{avg}\Delta t$$

$$v_{1f} = \frac{m_1 - m_2}{m_1 + m_2} v_{1i}$$

$$v_{2f} = \frac{2m_1}{m_1 + m_2} v_{1i}$$

$$\sin^2\theta + \cos^2\theta = 1$$

$$\sin 2\theta = 2\sin\theta\cos\theta$$

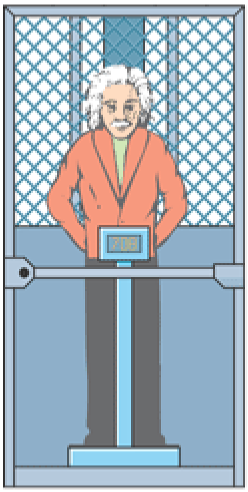
$$\sin\left(\alpha\pm\beta\right) = \sin\alpha\cos\beta \pm \cos\alpha\sin\beta$$

$$\cos\left(\alpha\pm\beta\right) = \cos\alpha\cos\beta \mp \sin\alpha\sin\beta$$

Question 1

In the figure, a passenger of mass m = 72.2 kg stands on a platform scale in an elevator cab. We are concerned with the scale readings when the cab is stationary and when it is moving up or down.

Die figuur dui 'n passasier, met massa m = 72.2 kg, aan wat in 'n hysbak op 'n platform-skaal staan. Ons stel belang in die lesing op die skaal wanneer die hysbak stilstaan en wanneer dit op en af beweeg.



(a)

(a) Draw the free body diagram and find a general solution for the scale reading, whatever the vertical motion of the elevator cab.
Teken die vry-liggaams diagram en bepaal die algemene oplossing vir die lesing op die skaal, ongeag die vertikale beweging van die hysbak.

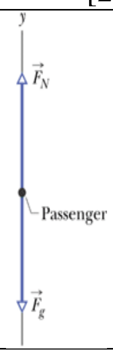
[2+2}

Because the two forces on the passenger and his acceleration are all directed vertically, along the y axis, we can use Newton's second law written for y components

$$\vec{F}_{net} = m\vec{a}$$

$$F_{net,y} = F_N - mg = ma$$

$$F_N = ma + mg = m(a+g)$$



(b) What does the scale read if the cab accelerates upward at 3.20 m/s^2?
Wat is die lesing op die skaal wanneer die hysbak opwaarts versnel teen 3.20 m/s^2?

[3]

$$a = 3.2 \text{ ms}^{-2}$$

$$F_N = m(a+g)$$
$$= 72.2(3.2+9.8)$$
$$= 939 \text{ N}$$

(c) During the upward acceleration in part (b), what is the magnitude F_{net} of the net force on the passenger, **AND** what is the magnitude of the passengers relative acceleration $a_{p,cab}$ as measured in the frame of the cab?
*Tydens die opwaartse versnelling in deel (b), wat is die grootte F_{net}, die netto krag, op die passasier, **EN** wat is die grootte van die passasier se relatiewe versnelling, $a_{p,cab}$, soos gemeet in die raamwerk van die hysbak?*

[4]

$$\vec{F}_{net} = m\vec{a} = 72.2(3.2) = 231 \text{ N}$$

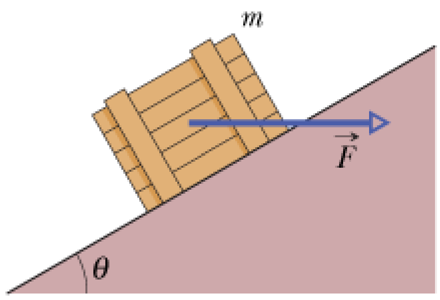
or

$$F_{net,y} = F_N - mg = 939 - (72.2(9.8)) = 231 \text{ N}$$

$$a_{p,cab} = 0 \text{ m/s}^2$$

Question 2

In the figure, a crate of mass m = 100 kg is pushed at constant speed up a frictionless ramp (θ = 30.0°) by a horizontal force $\vec{F}$.
Die figuur dui 'n krat met massa, m = 100 kg, aan wat teen 'n konstante spoed teen 'n wrywinglose skuinsvlak (θ = 30.0°) opgestoot word deur 'n horisontale krag $\vec{F}$.



(a) What is the magnitude of $\vec{F}$?
Wat is grootte van $\vec{F}$?

[3+2]

Newton's second law applied to the x-axis produces

$$F\cos\theta - mg\sin\theta = ma.$$

For $a = 0$, this yields $F = 566$ N.

(b) What is the magnitude of the force on the crate from the ramp?
Wat is die grootte van die krag op die krat wat deur die skuinsvlak daarop uitgeoefen word?

[3]

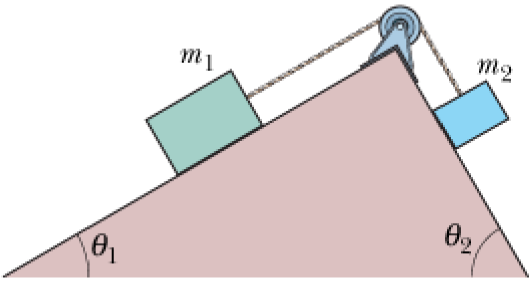
Applying Newton's second law to the y axis (where there is no acceleration), we have

$$F_N - F\sin\theta - mg\cos\theta = 0$$

which yields the normal force $F_N = 1.13 \times 10^3$ N.

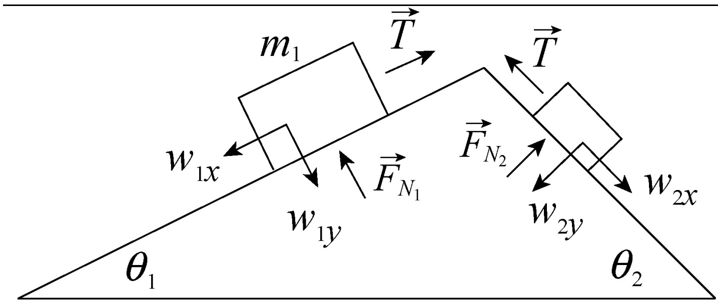
Question 3

The figure shows a box (mass m_1 = 3.0 kg) on a frictionless plane inclined at angle θ_1 = 30°. The box is connected via a cord of negligible mass to another box (mass m_2 = 2.0 kg) on a frictionless plane inclined at angle θ_2 = 60°. The pulley is frictionless and has negligible mass. Determine the tension in the cord.



Die figuur dui 'n boks (massa m_1 = 3.0 kg) op 'n wrywinglose skuinsoppervlak, θ_1 = 30°. Hierdie boks word deur 'n tou, van weglaatbare massa, met 'n ander boks (massa m_2 = 2.0 kg) verbind. Hierdie tweede boks lê ook op 'n wrywinglose skuinsoppervlak, θ_2 = 60°. Die katrol is wrywingloos, met weglaatbare massa. Bepaal die spanning in die koord.

[7]



Assume motion of the boxes is uphill for m_1 and downhill for m_2. x-axis is parallel to the slopes for the 2 masses. Motion is in the positive x direction for both boxes.

Newtons laws for the two masses

For mass 1

$$\sum F_y = N - m_1 g\cos\theta_1 = 0$$
$$\sum F_x = T - m_1 g\sin\theta_1 = m_1 a$$

For mass 2

$$\sum F_y = N - m_2 g\sin\theta_2 = 0$$
$$\sum F_x = -T + m_2 g\cos\theta_2 = m_2 a$$

Adding the two x component together:

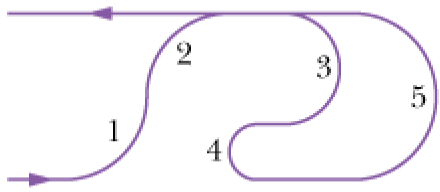
$$T - m_1 g\sin\theta_1 = m_1 a$$
$$m_2 g\sin\theta_2 - T = m_2 a$$

Solve for the acceleration:

$$a = \left(\frac{m_2\sin\theta_2 - m_1\sin\theta_1}{m_2 + m_1}\right) g$$

With $\theta_1 = 30°$ and $\theta_2 = 60°$, we have a = 0.45 m/s^2. This value is plugged back into either of the two equations to yield the tension T = 16 N.

Question 4



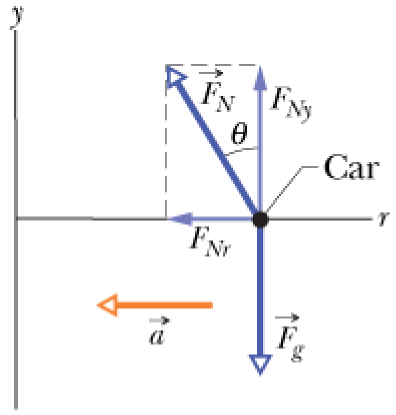
The figure shows the path of a park ride that travels at constant speed through five circular arcs of radii R_0, $2R_0$, and $3R_0$. Which of these arcs will produce the greatest centripetal force on a rider travelling in the arcs? Give reasons for your answer.
Die figuur dui die pad aan van 'n karnaval-rit wat teen 'n konstante snelheid deur vyf sirkelboë van radiusse R_0, $2R_0$, en $3R_0$ beweeg. Watter van die draaie sal die grootste sentripetale krag op die passasier veroorsaak? Gee redes vir jou antwoord.

[3]

4, a = v2/r, and if v is constant then the smaller the radius the higher the acceleration and the greater the force since F = ma.

Question 5

A banked circular highway curve is designed for traffic moving at 60 km/h. The radius of the curve is 200 m. Traffic is moving along the highway at 40 km/h on a rainy day.
'n Skuins sirkelvormige snelweg-draai word ontwerp vir verkeer wat teen 60 km/h beweeg. Die radius van die draai is 200 m. Verkeer beweeg daardeur op 'n reënerige dag teen 40 km/h.



(a) Determine the banking angle of the curve.
Bepaal die hoek van die skuins draai.

[6+2]

$$\sum F_x = N\sin\theta = m\left(\frac{v^2}{R}\right) \quad \ldots\ldots\ldots\ldots\ldots\ldots 1$$

$$\sum F_y = N\cos\theta - mg = 0 \quad \ldots\ldots\ldots\ldots\ldots\ldots\ldots 2$$

$$\therefore N\cos\theta = mg$$

1 / 2 gives:

$$\frac{N\sin\theta}{N\cos\theta} = \frac{m\left(\frac{v^2}{R}\right)}{mg}$$

$$\tan\theta = \frac{v^2}{gR}$$

$$\theta = \tan^{-1}\left(\frac{v^2}{gR}\right)$$

with v = 60(1000/3600) = 17 m/s

and R = 200 m. The banking angle is therefore θ = 8.1°.

(b) What is the minimum coefficient of friction between tires and road that will allow cars to take the turn without sliding off the wet road? Hint: Place your x-axis along the banked surface.
Wat is die minimum wrywingskoëffisiënt tussen die bande en die pad wat sal toelaat dat die motorkarre deur die draai beweeg sonder om van die nat pad af te gly? Wenk: Laat die x-as langs die skuinste lê.

[6+2]

Now we consider a vehicle taking this banked curve at v' = 40(1000/3600) = 11 m/s. Its (horizontal) acceleration is $a' = v'^2/R$, which has components parallel the incline and perpendicular to it:

$$a_{\parallel} = a'\cos\theta = \frac{v'^2\cos\theta}{R}$$

$$a_{\perp} = a'\sin\theta = \frac{v'^2\sin\theta}{R}.$$

These enter Newton's second law as follows (choosing downhill as the $+x$ direction and away-from-incline as $+y$):

$$mg\sin\theta - f_s = ma_{\parallel}$$

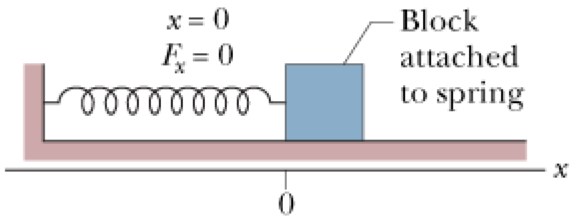
$$F_N - mg\cos\theta = ma_{\perp}$$

and we are led to

$$\frac{f_s}{F_N} = \frac{mg\sin\theta - mv'^2\cos\theta / R}{mg\cos\theta + mv'^2\sin\theta / R}.$$

We cancel the mass and plug in, obtaining f_s/F_N = 0.078. The problem implies we should set f_s = $f_{s,\max}$ so that, by Eq. 6-1, we have μ_s = 0.078.

Question 6



Derive the general expression to determine the work done by a massless, ideal spring as it moves a block of mass *m*, through a distance *d.* Start from the basic definition of work - $W = \vec{F}\cdot\vec{d} = Fd\cos\phi$. Assume that the block is on a frictionless surface.
Herlei die algemene vergelyking vir die bepaling van die arbeid verrig deur 'n massalose, ideale veer, soos dit 'n blok met massa m, deur 'n afstand d beweeg. Begin met die basiese definisie vir arbeid - $W = \vec{F}\cdot\vec{d} = Fd\cos\phi$. Veronderstel dat die blok op 'n wrywinglose oppervlak beweeg.

[7]

For a spring $\vec{F} = -k\vec{d}$ which is a variable force.

Cannot apply above formula for work directly – but divide the distance into small increments Δx over which F can be considered a constant.

The angle $\phi = 0$.

The sum of the work done over all the Δx will then give the total work, $W = \sum F\Delta x$.

In the limit $\Delta x \to 0$ we can write:

$W = \int_{x_i}^{x_f} F(x)\,dx$ then

$$W = \int_{x_i}^{x_f} F(x)\,dx$$
$$= \int_{x_i}^{x_f} (-kx)\,dx = -k\int_{x_i}^{x_f} (x)\,dx$$
$$= -k\left[\tfrac{1}{2}x^2\right]_{x_i}^{x_f}$$
$$W = \tfrac{1}{2}kx_{x_i}^2 - \tfrac{1}{2}kx_{x_f}^2$$

Question 7

Consider the following two springs: spring *A* is stiffer than spring *B* ($k_A > k_B$).
Beskou die volgende twee vere: Veer A is stywer as veer B ($k_A > k_B$).

(a) If the springs are compressed the same distance, which of two spring forces does more work? Give reasons for your answer.
Indien die twee vere saamgepers word oor dieselfde afstand, watter een van die twee veerkragte verrig meer arbeid? Gee redes vir jou antwoord.

[3]

Spring A: Work done by a spring: $W = -\tfrac{1}{2}kx^2$, if x is the same for both springs then k will determine the work done. Therefore the force due to Spring A will do more work.

If the springs are compressed by the same applied force, which of the two spring forces does more work? Give reasons for your answer.

Indien die twee vere saamgepers word deur dieselfde toegepaste krag, watter een van die twee veerkragte verrig meer arbeid? Gee redes vir jou antwoord.

[3]

Spring B:

Work done by the external force: $W = Fd$.

If F is the same for both springs, then Spring B will compress further because it is not so stiff, and therefore the work done by the spring = -Work done by the external force.

Question 8

The only force acting on a 2.0 kg body as it moves along a positive x axis has an x component $F_x = -6x\,\mathrm{N}$, with x in meters. The velocity at x = 3.0 m is 8.0 m/s.
Die enigste krag wat inwerk op 'n 2.0 kg liggaam soos dit langs die positiewe x-as beweeg het 'n x-komponent $F_x = -6x$ N, *met x in meter. Die snelheid by x = 3.0 m is 8.0 m/s.*

(a) What is the velocity of the body at x = 4.0 m?
Wat is die snelheid van die liggaam by x = 4.0 m?

[4]

As the body moves along the x axis from x_i = 3.0 m to x_f = 4.0 m the work done by the force is

$$W = \int_{x_i}^{x_f} F_x\,dx = \int_{x_i}^{x_f} -6x\,dx = -3(x_f^2 - x_i^2) = -3\,(4.0^2 - 3.0^2) = -21\text{ J}.$$

According to the work-kinetic energy theorem, this gives the change in the kinetic energy:

$$W = \Delta K = \frac{1}{2}m\left(v_f^2 - v_i^2\right)$$

where v_i is the initial velocity (at x_i) and v_f is the final velocity (at x_f). The theorem yields

$$v_f = \sqrt{\frac{2W}{m} + v_i^2} = \sqrt{\frac{2(-21\text{ J})}{2.0\text{ kg}} + (8.0\text{ m/s})^2} = 6.6\text{ m/s}.$$

(b) At what positive value of x will the body have a velocity of 5.0 m/s?
By watter positiewe waarde van x sal die liggaam 'n snelheid van 5.0 m/s hê?

[3]

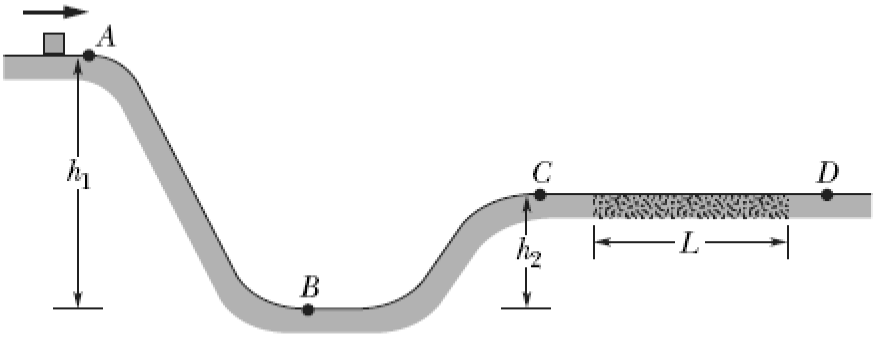
The velocity of the particle is v_f = 5.0 m/s when it is at $x = x_f$. The work-kinetic energy theorem is used to solve for x_f. The net work done on the particle is $W = -3\left(x_f^2 - x_i^2\right)$, so the theorem leads to

$$-3\left(x_f^2 - x_i^2\right) = \frac{1}{2}m\left(v_f^2 - v_i^2\right).$$

Thus,

$$x_f = \sqrt{-\frac{m}{6}\left(v_f^2 - v_i^2\right) + x_i^2} = \sqrt{-\frac{2.0\text{ kg}}{6\text{ N/m}}\left((5.0\text{ m/s})^2 - (8.0\text{ m/s})^2\right) + (3.0\text{ m})^2} = 4.7\text{ m}.$$

Question 9



In the figure, a small block is sent through point *A* with a speed of 7.0 m/s. Its path is without friction until it reaches the section of length L = 12 m, where the coefficient of kinetic friction is 0.70. The indicated heights are h_1 = 6.0 m and h_2 = 2.0 m.

In die figuur, 'n klein blokkie word deur punt A gestuur teen 'n spoed van 7.0 m/s. Die pad wat gevolg word is wrywingloos tot die lengte L = 12 m, waar die koefisient van wrywing 0.70 is. Die hoogtes wat aangedui word is h_1 = 6.0 m en h_2 = 2.0 m.

a) Without using the equations of motion for constant acceleration, determine the speed of the block at point *B*?
Sonder om die bewegings vergelykings vir konstante versnelling te gebruik bepaal die spoed van die blokkie by punt B?

[3]

$$\Delta K = \Delta U$$
$$K_B - K_A = U_B - U_A$$
$$½mv_B^2 - ½mv_A^2 = 0 - mgh_1$$
$$v = \sqrt{v_A^2 + 2gh_1}$$
$$= \sqrt{(7.0)^2 + 2\left((9.8\ \text{m/s}^2\right)(6.0\ \text{m})}$$
$$= 13\ \text{m/s}$$

If equations of motion were used and the answer was correct then award ***0*** *marks*

b) Without using the equations of motion for constant acceleration, determine the speed of the block at point C ?
Sonder om die bewegings vergelykings vir konstante versnelling te gebruik bepaal die spoed van die blokkie by punt C?

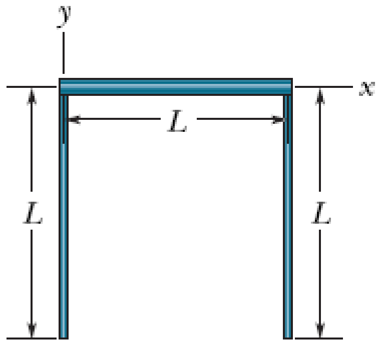
[3]

$$\Delta K = \Delta U$$
$$K_C - K_A = U_C - U_A$$
$$½mv_C^2 - ½mv_A^2 = 0 - mgh_1$$
$$v_C = \sqrt{v_A^2 + 2gh_1}$$
$$= \sqrt{(7.0)^2 + 2\left((9.8\ \text{m/s}^2\right)(4.0\ \text{m})}$$
$$= 11.3\ \text{m/s}$$

If equations of motion were used and the answer was correct then award ***0*** *marks*

Question 10

In the figure, three uniform thin rods, each of length L = 22 cm, form an inverted U. The vertical rods each have a mass of 14 g; the horizontal rod has a mass of 42 g. What are the x- and y-coordinates of the system's centre of mass? Hint: Use the centre of mass of each rod.
Die figuur dui drie univorme dun stafies aan, elk met lengte L = 22cm, wat 'n omgekeerde U vorm. Die vertikale stafies het elk 'n massa van 14 g, en die vertikale stafie het 'n massa van 42 g. Bepaal die x- en y-koordinate van die stelsel se swaartepunt. Wenk: Gebruik die swaartepunt van elk een van die stafies.



[5]

With the origin of the axes located at the top left side of the arrangement, we continue.
Com of LHS rod: x = 0 cm $y = -L/2$ cm
Com of RHS rod: x = L cm $y = -L/2$ cm
Com of top rod: x = L/2 cm y = 0 cm

The x coordinate of the assembly's center of mass is

$$x_{\text{com}} = \frac{M(+L) + M(0) + 3M(+L/2)}{M + M + 3M} = 0.5L\,.$$

With L = 22 cm, we have x_{com} = 11 cm.

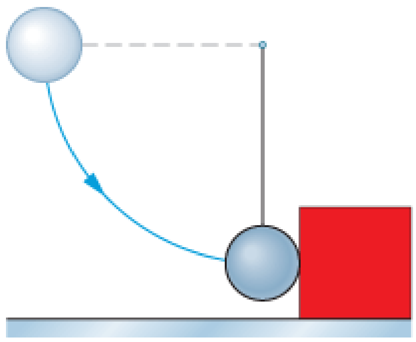
The y coordinate of the system's center of mass is

$$y_{\text{com}} = \frac{M(-L/2) + M(-L/2) + 3M(0)}{M + M + 3M} = -\frac{L}{5},$$

or $y_{\text{com}} = -4.4$ cm.

Question 11

A steel ball of mass 0.500 kg is fastened to a cord that is 70.0 cm long and fixed at the far end. The ball is then released when the cord is horizontal (As shown in the figure). At the bottom of its path, the ball strikes a 2.50 kg steel block initially at rest on a frictionless surface. The collision is elastic. Find the following:
'n Staal bal, met massa 0.500 kg is vasgemaak aan 'n tou wat 70 cm lank is, en dit is vasgemaak aan die een kant. Die bal word gelos wanneer die tou horisontaal is (Soos aangedui in die figuur). Wanneer die bal onder kom, slaan dit teen 'n aanvanklik stilstaande, 2.50 kg staal blok vas wat op 'n wrywinglose oppervlak staan. Die botsing is elasties. Bereken die volgende:



(a) the speed of the ball just after the collision.
die spoed van die bal net na die botsing.

[5]

Speed v of the ball of mass m_1 right before the collision (just as it reaches its lowest point of swing). Mechanical energy conservation (with h = 0.700 m) leads to

$$m_1 gh = \frac{1}{2} m_1 v^2 \;\Rightarrow\; v = \sqrt{2gh} = 3.7 \text{ m/s}.$$

We now treat the elastic collision:

$$\begin{aligned} v_{1f} &= \frac{m_1 - m_2}{m_1 + m_2} v_{1i} \\ &= \frac{0.5 \text{ kg} - 2.5 \text{ kg}}{0.5 \text{ kg} + 2.5 \text{ kg}} (3.7 \text{ m/s}) \\ &= -2.47 \text{ m/s} \end{aligned}$$

which means the final speed of the ball is 2.47 m/s.

(b) the speed of the block just after the collision.
spoed van die blok net na die botsing.

[3]

find the final speed of the block:

$$v_{2f} = \frac{2m_1}{m_1 + m_2} v = \frac{2(0.5 \text{ kg})}{0.5 \text{ kg} + 2.5 \text{ kg}} (3.7 \text{ m/s}) = 1.23 \text{ m/s}.$$

Goodbye!
Totsiens!